

EE473 - HOMEWORK 5

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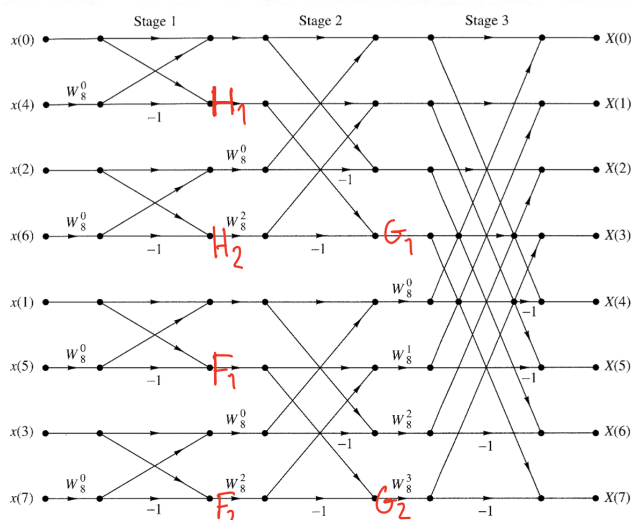
1) a) $\text{Gain} = W_8^0 W_8^0 (-1) W_8^2 = 1 \times 1 \times (-1) \times e^{-j\pi/2} = \boxed{j}$

↳ $X(7)$ will follow the path "up" + "straight" + "up" through butterflies.

b) There is only one path from an input to a given output.

↳ In fact theoretically, the gain W_N^{ij} represents the gain of the path from input j to output i .

c)



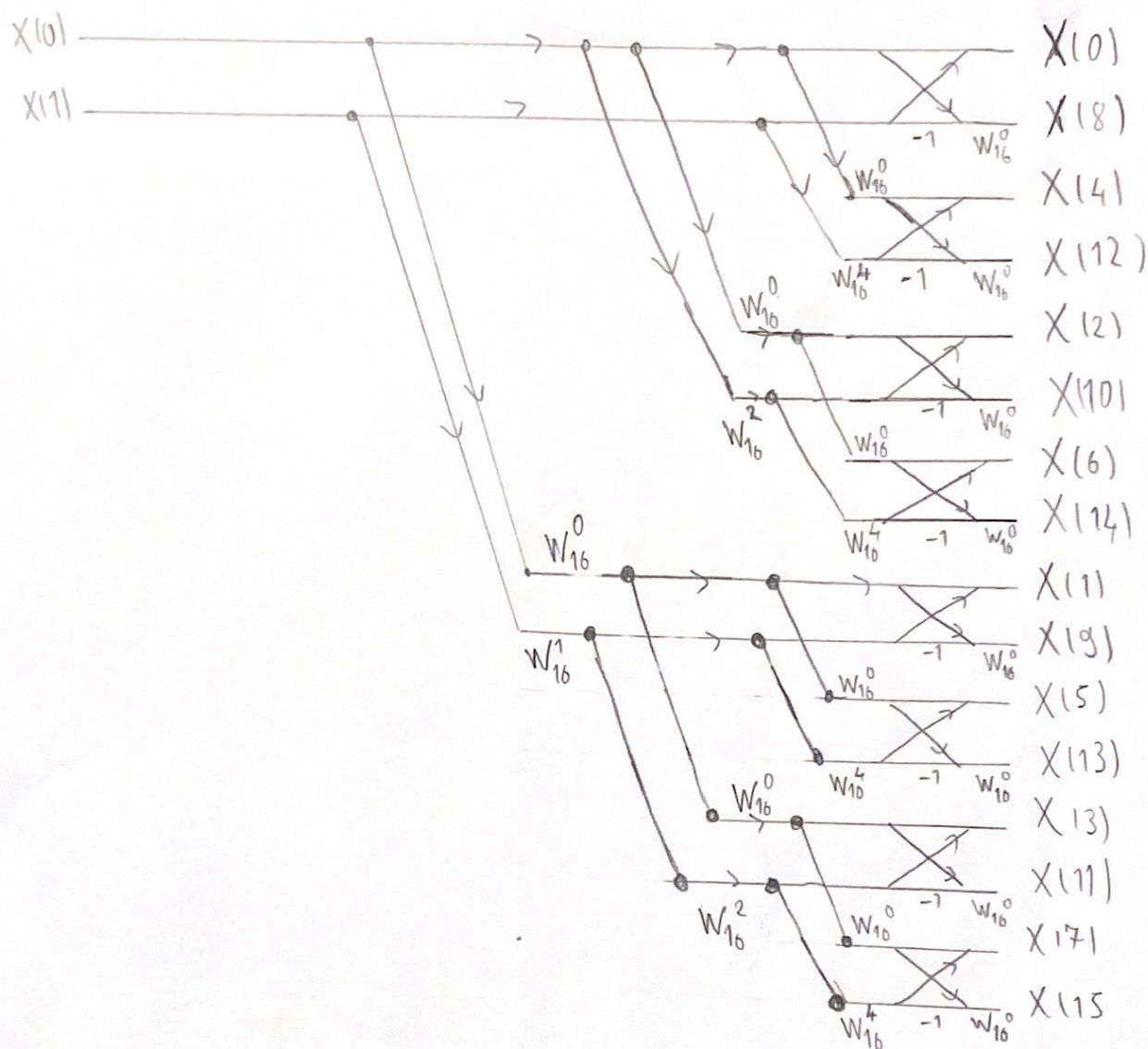
$$\begin{aligned} X(3) &= G_1 + W_8^3 G_2 \\ &= H_1 - W_8^2 H_2 + W_8^3 (F_1 - W_8^2 F_2) \\ &= \underbrace{X(0) - W_8^0 X(4)}_{H_1} - W_8^2 \underbrace{(X(2) - W_8^0 X(6))}_{H_2} + W_8^3 \underbrace{(X(1) - W_8^0 X(5))}_{F_1} \\ &\quad - W_8^3 W_8^2 \underbrace{(X(3) - W_8^0 X(7))}_{F_2} \end{aligned}$$

⇒ Noting $W_8^0 = 1$

$$\begin{aligned} X(3) &= X(0) + W_8^3 X(1) - W_8^2 X(2) - W_8^3 W_8^2 X(3) - X(4) - W_8^3 X(5) + W_8^2 X(6) + W_8^2 W_8^3 X(7) \\ &= X(0) + e^{j3\pi/4} X(1) + j X(2) + e^{j\pi/4} X(3) - X(4) + e^{-j\pi/4} X(5) - j X(6) + e^{-j3\pi/4} X(7) \end{aligned}$$

2)

a)



* Ordering of $X(k)$'s are determined by bit reversal

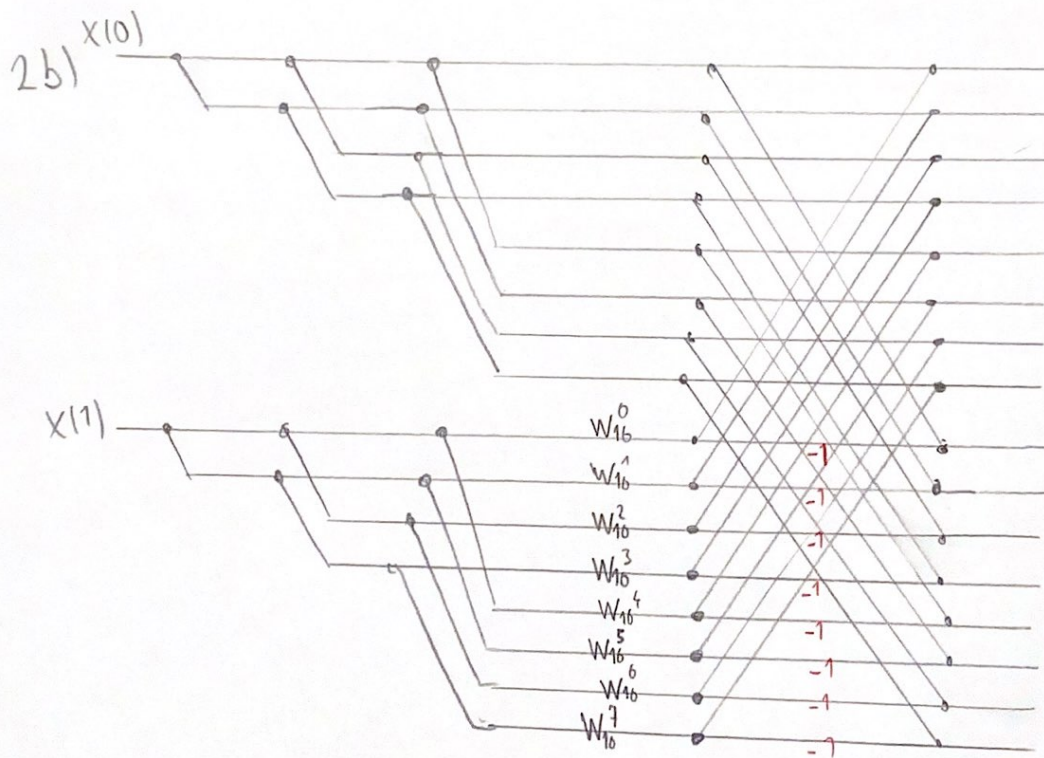
c) For the input pruning cases, DIF is more advantageous since we are only doing complex multiplications with trivial coeffs. $W_{16}^0, W_{16}^1, W_{16}^2, W_{16}^4$ compared to $W_{16}^0, \dots, W_{16}^7$ in DIT.

Even when $X(0), X(1), X(2), X(3)$ are to be calculated, we only need to do a few mult. w/ W_{16}^1, W_{16}^2 compared to $W_{16}^1, W_{16}^2, W_{16}^3$ which is more difficult. So DIF is still better

2.c) In the heuristic case where we compute M points of the DFT from L points of the input sequence,

↳ For sparse input ($L \ll N$) (or input pruning) DIF is better since it eliminates the early complex mult.

↳ For Sparse output ($M \ll N$) (or output pruning) DIT is better since it allows you to skip the final complex mult.



2d) In classical FFT, we have $\log_2 N$ stages and each stage requires $\frac{N}{2}$ butterflies (1 mult + 2 add) $\Rightarrow C_{FFT} = \frac{N}{2} \log_2 N$ complex mult.

In pruned FFT where input and output is very sparse, we'll have approximately $M \times L$ input-output pairs. This requires $\frac{ML}{2}$ butterflies per stage in FFT and we have again $\log_2 N$ stages $\Rightarrow C_{prune} = \frac{ML}{2} \log_2 N$

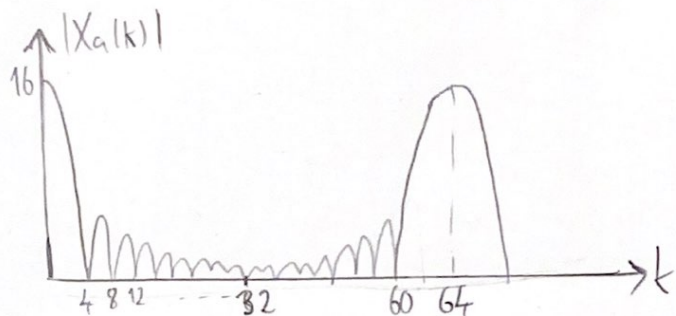
$$\% \text{ Saving} = \frac{\frac{N}{2} \log_2 N - \frac{ML}{2} \log_2 N}{\frac{N}{2} \log_2 N} \times 100\% = \left(1 - \frac{ML}{N}\right) \times 100\%$$

3) N-point DFT of a rectangular window of size N_1 is :

$$X_N^{(N_1)}(k) = \sum_{n=0}^{N_1-1} 1 \cdot e^{-j2\pi kn/N} = \frac{1 - e^{-j2\pi k N_1/N}}{1 - e^{-j2\pi k/N}} = e^{-j\pi k(N_1-1)/N} \cdot \frac{\sin(\pi k N_1/N)}{\sin(\pi k/N)}$$

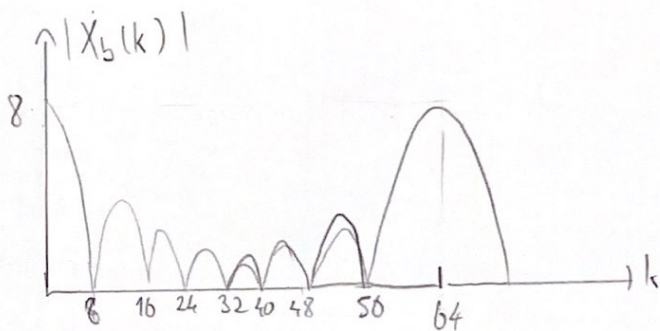
a) $x_a(n) = \begin{cases} 1, & n=0, 1, \dots, 15 \\ 0, & \text{o.w.} \end{cases}$

$\hookrightarrow X_a(k) = e^{-j\pi k \frac{15}{64}} \frac{\sin(\pi k/4)}{\sin(\pi k/64)}$
 $\hookrightarrow$ 64-point DFT



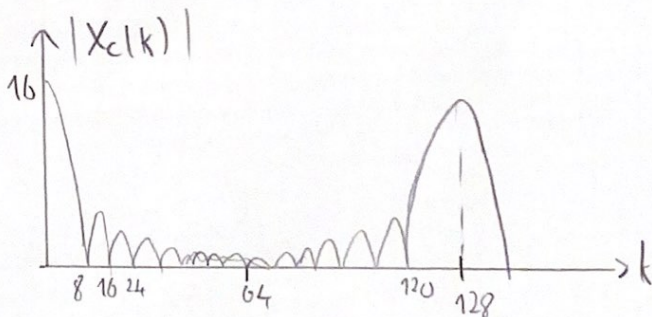
b) $x_b(n) = \begin{cases} 1, & n=0, 1, \dots, 7 \\ 0, & \text{o.w.} \end{cases}$

$\hookrightarrow X_b(k) = e^{-j\pi k \frac{7}{64}} \frac{\sin(\pi k/8)}{\sin(\pi k/64)}$
 $\hookrightarrow$ 64-point DFT



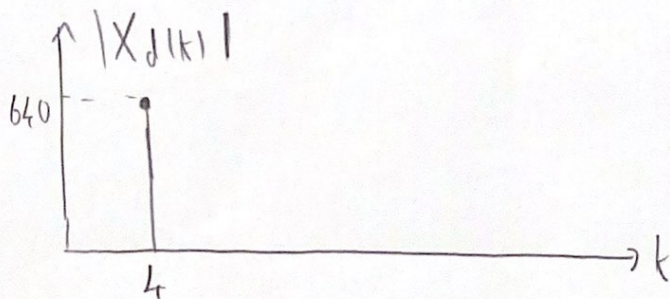
c) $x_c(n) = \begin{cases} 1, & n=0, \dots, 15 \\ 0, & \text{o.w.} \end{cases}$

$\hookrightarrow X_c(k) = e^{-j\pi k \frac{15}{128}} \frac{\sin(\pi k/8)}{\sin(\pi k/128)}$



d) $x_d(n) = \begin{cases} 10e^{j\pi/8 n}, & n=0, \dots, 63 \\ 0, & \text{o.w.} \end{cases}$

$X_d(k) = \sum_{n=0}^{63} 10e^{j2\pi n \frac{(k-4)}{64}}$
 $= 640 \delta(k-4)$



1) Frequency interval is $\frac{1}{N}$ where N is number of points in DFT

a) $\Delta f_a = \frac{1}{64}$ b) $\Delta f_b = \frac{1}{64}$ c) $\Delta f_c = \frac{1}{128}$ d) $\Delta f_d = \frac{1}{64}$

$\rightarrow$ Or $\frac{2\pi}{N}$ if one want to look at ω .

$$2) \quad X(0) = \sum_{n=0}^{N-1} x(n) \cdot 1$$

$$X_a(0) = 16, \quad X_b(0) = 8, \quad X_c(0) = 16, \quad X_d(0) = 0$$

↳ All results match with the values of DFTs plotted at $k=0$

$$3) \quad a) \quad \Delta k_{\text{null}} = 4 \quad b) \quad \Delta k_{\text{null}} = 8 \quad c) \quad \Delta k_{\text{null}} = 8$$

$$\Delta f_{\text{null}} = \frac{1}{16} \quad \Delta f_{\text{null}} = \frac{1}{8} \quad \Delta f_{\text{null}} = \frac{1}{16}$$

$$\Delta \omega_{\text{null}} = \frac{\pi}{8} \quad \Delta \omega_{\text{null}} = \frac{\pi}{4} \quad \Delta \omega_{\text{null}} = \frac{\pi}{8}$$

$$\text{↳ Nulls occur with } \Delta k = \frac{N}{N_1}, \quad \Delta f = \frac{1}{N_1}, \quad \Delta \omega = \frac{2\pi}{N_1}$$

4) 128-point DFT^(c) has twice as many samples compared to (a). Thus (c) has better resolution than (a) although the shape of the envelope is the same.

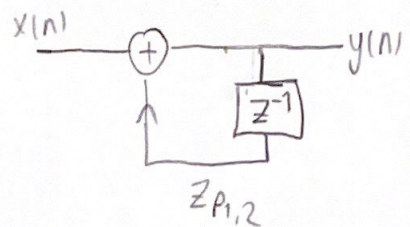
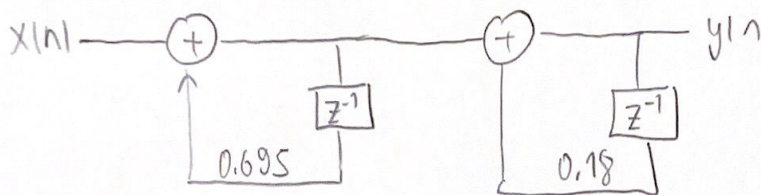
$$4) \quad y(n) = 0.875 y(n-1) - 0.125 y(n-2) + x(n)$$

$$a) \quad H(z) = \frac{1}{0.125(8 - 7z^{-1} + z^{-2})} \rightarrow z_{p1} = \frac{7 + \sqrt{49 - 32}}{16} = 0.180$$

$$\rightarrow z_{p2} = \frac{7 - \sqrt{49 - 32}}{16} = 0.695$$

$$\underbrace{(1 - z_{p1}z^{-1})}_{G_1(z)} \underbrace{(1 - z_{p2}z^{-1})}_{G_2(z)}$$

We can realize $G_{1,2}(z) = 1 - z_{p1,2}z^{-1}$ by

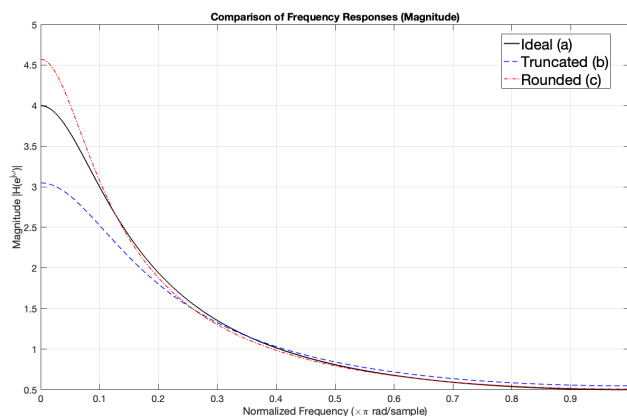


b) By quantizing coeffs., locations of the poles are understood since 0.875 and 0.125 are already 3 bits.

$$\left. \begin{aligned} z_{p1} = 0.695 &\rightarrow \frac{5}{8} = 0.625 : 101 \\ z_{p2} = 0.180 &\rightarrow \frac{1}{8} = 0.125 : 001 \end{aligned} \right\} \begin{cases} \tilde{z}_{p1} = 0.625 \\ \tilde{z}_{p2} = 0.125 \end{cases}$$

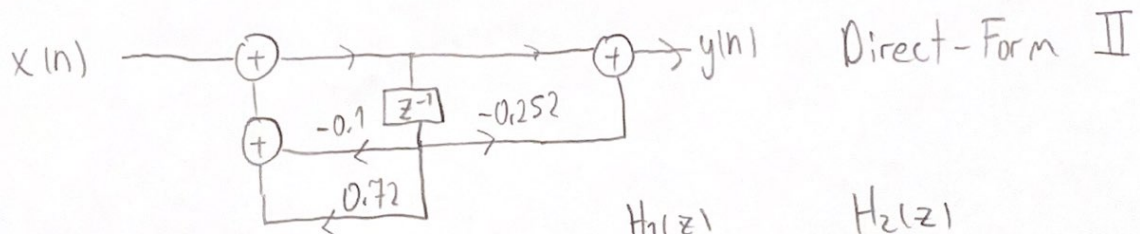
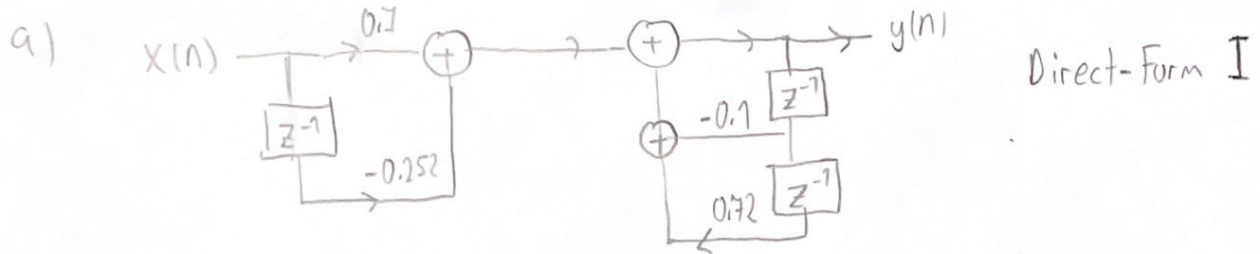
$$\left. \begin{aligned} z_{p1} = 0.695 &\simeq \frac{6}{8} = 0.750 : 110 \\ z_{p2} = 0.180 &\simeq \frac{1}{8} = 0.125 : 001 \end{aligned} \right\} \begin{cases} \hat{z}_{p1} = 0.750 \\ \hat{z}_{p2} = 0.125 \end{cases}$$

d) 0.75 is closer to 0.695 than 0.625, thus rounding is better

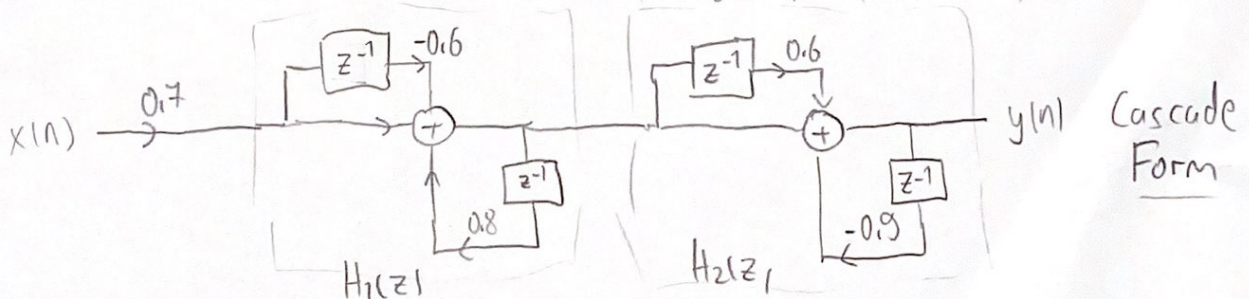


* Rounding curve (red) is closer to ideal case compared to trunc.

5) $y[n] = -0.1 y[n-1] + 0.72 y[n-2] + 0.7 x[n] - 0.252 x[n-2]$



$$H(z) = \frac{0.7 - 0.252z^{-2}}{1 + 0.1z^{-1} - 0.72z^{-2}} \approx 0.7 \frac{(1 - 0.6z^{-1})}{(1 - 0.8z^{-1})} \cdot \frac{(1 + 0.6z^{-1})}{(1 + 0.9z^{-1})}$$



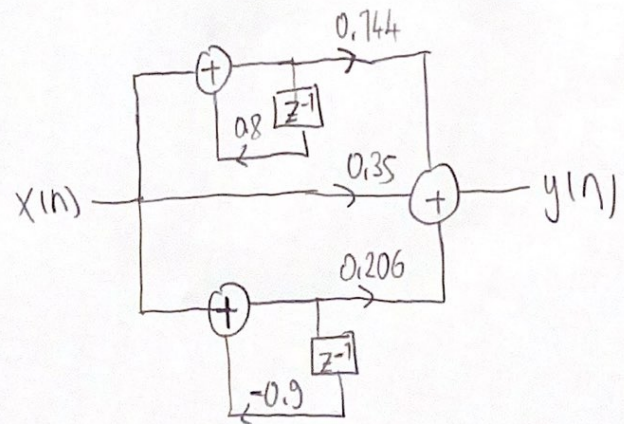
$$\frac{H(z)}{z} = \frac{C}{z} + \frac{A}{z-0.8} + \frac{B}{z+0.9}$$

$$C = \left. \frac{0.7z^2 - 0.252}{(z-0.8)(z+0.9)} \right|_{z=0} = 0.35$$

$$A = \left. \frac{0.7z^2 - 0.252}{z(z+0.9)} \right|_{z=0.8} \approx 0.144$$

$$B = \left. \frac{0.7z^2 - 0.252}{z(z-0.8)} \right|_{z=-0.9} \approx 0.206$$

$$H(z) = 0.35 + \frac{0.144}{1 - 0.8z^{-1}} + \frac{0.206}{1 + 0.9z^{-1}}$$



5b)

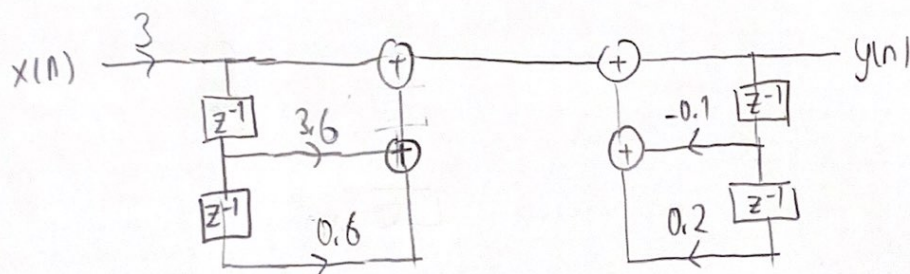
$$y(n) = -0.1y(n-1) + 0.2y(n-2) + 3x(n) + 3.6x(n-1) + 0.6x(n-2)$$

$$H(z) = \frac{3(1 + 1.2z^{-1} + 0.2z^{-2})}{1 + 0.1z^{-1} - 0.2z^{-2}} = 3 \frac{(1+z^{-1})(1+0.2z^{-1})}{\underbrace{(1-0.4z^{-1})}_{H_1(z)} \underbrace{(1+0.5z^{-1})}_{H_2(z)}}$$

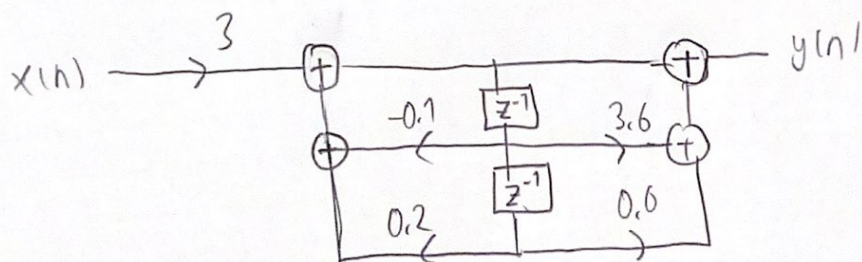
$$H(z) = -3 + \frac{6 + 3.9z^{-1}}{(1-0.4z^{-1})(1+0.5z^{-1})} = -3 + \frac{A}{1-0.4z^{-1}} + \frac{B}{1+0.5z^{-1}}$$

$$A = \left. \frac{6 + 3.9z^{-1}}{1 + 0.5z^{-1}} \right|_{z=0.4} = 7, \quad B = \left. \frac{6 + 3.9z^{-1}}{1 - 0.4z^{-1}} \right|_{z=0.5} = -1$$

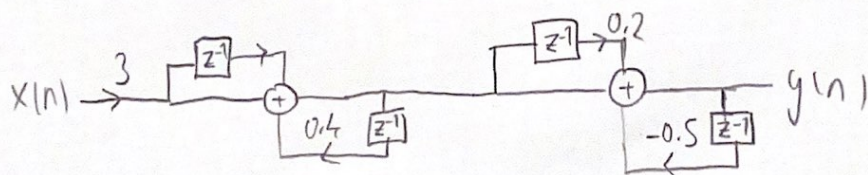
$$H(z) = -3 + \frac{7}{1-0.4z^{-1}} - \frac{1}{1+0.5z^{-1}}$$



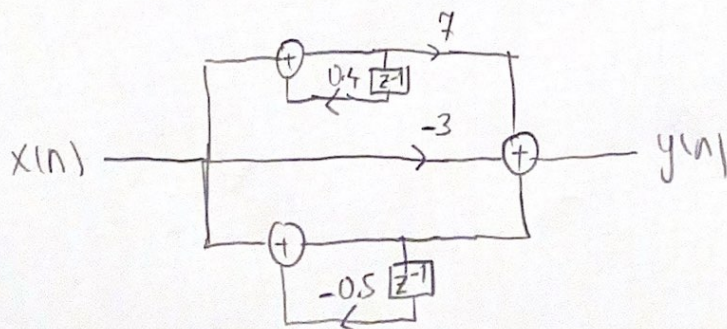
Direct-Form I



Direct-Form II



Cascade Form



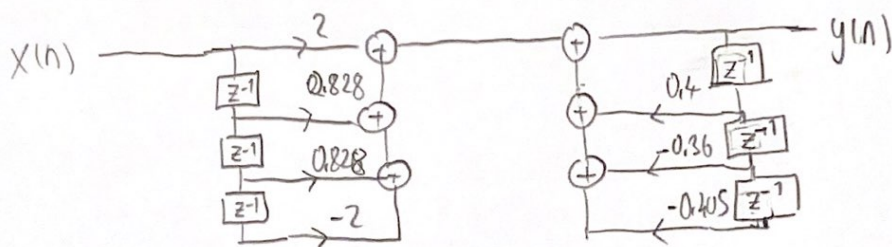
Parallel Form

$$5c) H(z) = \frac{2(1-z^{-1})(1+\sqrt{2}z^{-1}+z^{-2})}{(1+0.5z^{-1})(1-0.9z^{-1}+0.81z^{-2})}$$

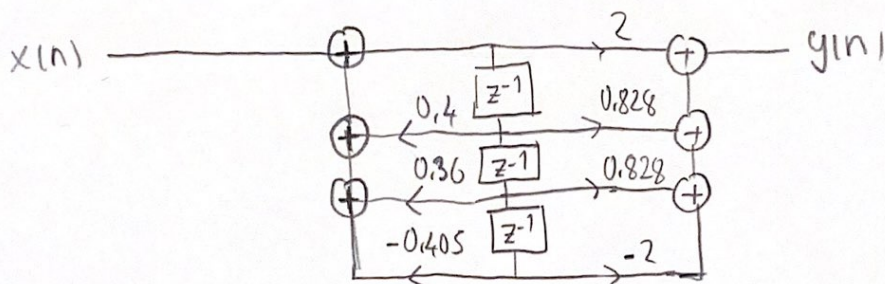
$$H(z) \cong -4.94 + \frac{2.157}{1+0.5z^{-1}} + \frac{4.78-1.6z^{-1}}{1-0.9z^{-1}+0.81z^{-2}}$$

(computed via MATLAB)

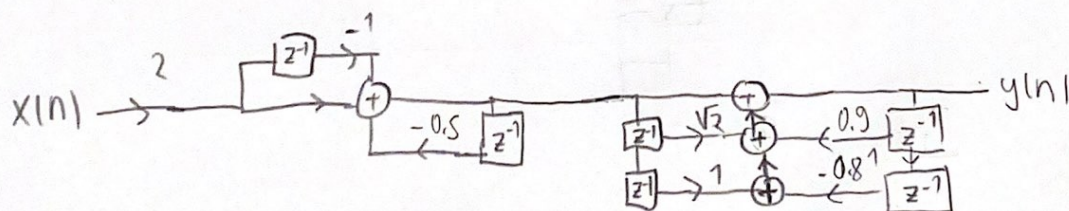
$$H(z) = \frac{2 + 0.828z^{-1} - 0.828z^{-2} - 2z^{-3}}{1 - 0.4z^{-1} + 0.36z^{-2} + 0.405z^{-3}}$$



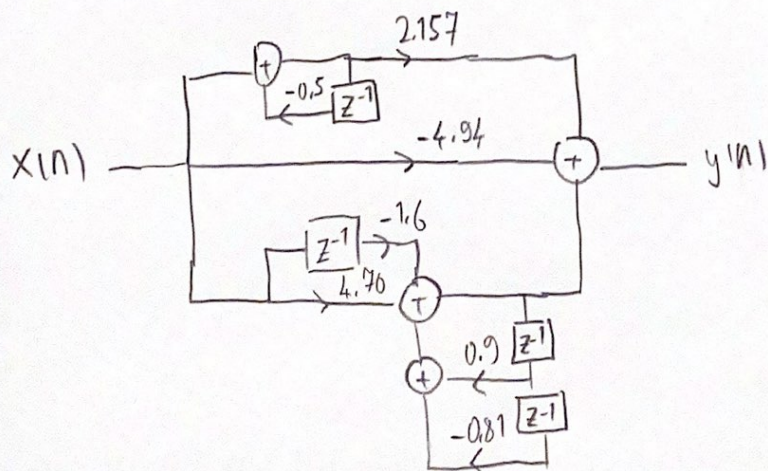
Direct-Form I



Direct-Form II



Cascade Form



Parallel Form